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RESEARCH ARTICLE

Modeling predictive suitability to identify the potential of wind and solar energy as a driver of sustainable development in the Red Sea state, Sudan

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Abstract

This paper investigated the potential and economic validity of wind and solar energy at 17 selected locations in the Red Sea state, Sudan, for the first time. To this aim, the NASA database was utilized. The results demonstrated that vertical axis wind turbines would be a good solution for electricity generation for building in the selected locations. Additionally, it is found that the chosen areas are suitable for installing photovoltaic (PV) systems due to the high-value solar radiation. Moreover, the economic viability of small-scale wind and PV systems for rooftop buildings in the selected regions is investigated. For a financial analysis of wind turbines, the performance of different characteristics of vertical axis winds was evaluated based on the determination of capacity factor and energy production cost. For the economic validity of installing PV systems, RETScreen Expert software was used. The results indicate that the annual production energy from wind turbines and solar power is within the range of 158.50–29,063.93kWh and 6648–15,533 kWh, respectively. This amount of energy output would reduce the effect of global warming and enhance the sustainable technological development of the country. Moreover, the results indicate that model#9 (Vertical Axis Wind Generator-V) with a capacity of 5 kW has the lowest cost value (0.08703–0.01025 $/kWh) compared to the other selected turbines for the studied locations. Besides, the average energy production cost is within the range of 0.036–0.049 $/kWh for PV systems. In the end, it is concluded that using small-scale renewable energy systems will help reduce the electricity bills and the dependency on fossil fuels, the effect of global warming, and enhance the country's sustainable technological development.

Keywords Red Sea state · Sudan · Grid-connected · Rooftop PV system · RETScreen software · Techno-economic evaluation

Introduction

The high electricity consumption has motivated the researchers to find an alternative energy source to reduce energy and environmental impacts due to fossil fuels (Riahi et al. 2017; Vijayavenkataraman et al. 2012; Arreyndip and Joseph 2018). Several researchers concluded that using renewable energy like solar and wind energy as power sources helped reduce greenhouse gas (GHG) emissions (Schnitzer et al. 2007; Shahsavari and Akbari 2018; Shahsavari et al. 2019). Moreover, several scientific researchers investigated the potential and the economic feasibility of renewable energy in various countries around the world (Adnan et al. 2012; Kassem et al. 2019a, b; Enongene et al. 2019; Chauhan and Saini 2016; Himri et al. 2020; Ahmed 2021; Kassem et al. 2020a). For instance, Chauhan and Saini (2016) investigated an integrated renewable energy system (biogas, biomass, wind,

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and solar) in the Chamoli district of Uttarakhand state, India. It was found that the most suitable option for the selected location was a micro-hydropower-biogas-biomass-wind-PV-array-battery-based configuration. Himri et al. (2020) evaluated wind energy's potential and economic viability in Algeria's South-West region. The results indicated that the project was economically viable with good energy production and capacity factor. Ahmed (2021) investigated the techno-economic feasibility of wind farms in Sidi Barrani province, Egypt. The author concluded that a developed wind farm could produce approximately 988 GWh/year with an electricity production cost of 1.7 USC/kWh. Kassem et al. (2019a, b) evaluated solar and wind energy's economic viability and potential to meet the electricity demand for a small household in Northern Cyprus. It was found that the most suitable option for the selected location was a solar PV system. Enongene et al. (2019) investigated the economic viability of small-scale PV systems in Nigeria. They found that the developed system could reduce GHG emissions (mainly CO_2).

Electrical energy in Sudan is currently produced by thermal power and hydro-power plants (see Table S1 as supplementary material) according to Sudanese Electricity Distribution Company Ltd. Sudan's electric power sector has a substantial generation shortfall, resulting in frequent power outages; as a result, the electricity demand outweighs the present supply. Most of the regions, such as the Red Sea state, are suffering from power outages in the summer season, particularly Portsudan city, an industrial town, and needs regular electrical stability that has severe power outages, over six hours, during the daytime due to the schedule of the Sudanese Electricity Distribution Company Ltd (SEDC). Due to this reason, the residential is using generators as power sources during the outage of power. Additionally, the West of Sudan includes Darfur states, Al Fashir city, Nyala city, Al Junaynah city, Kordofan states, El Obeid City, and El Fula city; Kaduqli city and so on are suffering from outage of power. Furthermore, the national electrical network does not cover some regions, mainly rural areas. Thus, these regions rely on small-diesel fuel power generators. Besides, the industrial sector suffers from the electric current's instability because the current's instability affects the production process costs. As a result, most commercial factory owners rely on diesel generators to provide electricity during power outages.

Therefore, utilizing renewable energies such as wind and solar energy as power sources can be an alternative solution for solving the electricity crises in the country. Several scientific researchers have investigated the potential of solar and wind energy as a power generation source and in different locations in Sudan (Omer 1997, 2000, 2001, 2002, 2007a, b, 2008a, b; Omer and Braima 2007; Omee 2013; Salih et al. 2014; Salih and Aydrous 2015; Khadam and Duod 2017; Elzubeir 2016; Langodan et al. 2016; Ali 2018; Elkadeem et al. 2019; Osman and Sevinc 2019; Abdeen et al. 2019; Mubark Saeed et al. 2019; Hamza and Abdelatif 2020; Fadlallah and Serradj 2020; Fadlallah et al. 2021; Abdalla and Özcan 2021). For example, Omer (2007a) concluded that Sudan has huge solar energy potential due to sunshine and solar radiation and moderate wind speeds. Khadam and Duod (2017) investigated the wind energy potential as a power source for remote areas in Sudan using windPRO software. The results showed that the highest wind potential is recorded in Red Sea Coast and the Northern States compared to other Sudanese regions. Abdeen et al. (2019) developed a 5-MW grid-connected PV system in ten locations in Sudan. The results showed that the proposed system annually produced 8.56GWh. Abdalla and Özcan (2021) developed a 1-GW solar PV plant in the northern part of Sudan using PVsyst software. They found that the developed system is considered economically, technically, and environmentally feasible in Sudan.

Based on the findings, it is concluded that (1) utilizing the solar power system in Sudan is limited to simple applications like water pumping systems for irrigation in agriculture; (2) the hybrid system (diesel fuel-renewable energy) would help to reduce fuel consumption and lead to be more sustainable development; (3) solar power system could solve the electricity crisis and reduce the CO_2 emissions in the country; and (4) no study investigates the potential and the viability of small-scale wind and solar energy in the Red Sea state in Sudan.

Moreover, the findings demonstrated an evident lack of utilizing wind and solar systems for households in Sudan. In addition, according to the authors' review, no studies are evaluating the feasibility of grid-connected wind and PV systems as power sources for the building in the country. Therefore, the study aims to investigate the economic viability and environmental sustainability of grid-connected wind and PV systems in 17 selected regions in the Red Sea state of Sudan. For this purpose, the National Aeronautics and Space Administration (NASA) database as a source of meteorological information has been utilized to find a suitable region for installing the wind and PV systems in the selected state. In the present study, the technical electricity generation assessment and economic analysis of wind energy turbines are examined based on the present value cost (PVC) method. Besides, RETScreen software is used to analyze the technical, financial, economic, and environmental analysis of the PV system.

Material and methods

The solar and wind energy potential in the Red sea state, Sudan, is investigated in this section. In addition, the economic validity of grid-connected wind and solar systems is examined. Figure 1 shows the schematic diagram of the developed methodology in this study.

Data and study area

Sudan (see Fig. 2), with an area of 1.8 million square kilometers, is the third-largest country in Africa. The length of the coast of Sudan is 853 m. It has a tropical climate with large daily and seasonal temperature changes. The maximum temperature in summer ranges from 37 to 44 °C, and between 18 and 21 °C in the coldest months. The climatic zones range from arid desert in the north to semi-arid in the far south, influenced by the monsoons. The geographical coordinates and altitude of the selected locations are listed in Table 1.

In general, several studies evaluated the wind energy potential (Arreyndip et al. 2016; Rafique et al. 2018; Gökçekuş et al. 2019) and solar energy potential (Kassem et al. 2018, 2020b, 2021a; Owolabi et al. 2019; Kassem et al. 2019a, b) in different locations using the NASA database. Moreover, the NASA database showed a good correlation with measured data (global solar irradiation) according to Kassem et al. (2020c), Belkilani et al. (2018), and Gairaa and Bakelli (2013). Therefore, for evaluating the wind and solar energy potential, NASA average monthly wind and solar data provided by NASA POWER datasets are utilized.

Procedure of wind energy analysis

Distribution functions

Several scientific researchers have studied the wind speed characteristics at the specific location (Alayat et al. 2018; Kassem et al. 2019a, b; Khan et al.2019; Alavi et al.2016; Masseran 2015; Ouarda et al. 2015; Kassem et al. 2021b). The most used distribution functions in the literature are Weibull, Gamma, Lognormal, Logistic, Log-Logistic, Inverse Gaussian, Generalized Extreme Value, Nakagami, Normal, Rayleigh, generalized Gamma distribution, exponential, and Kappa distribution functions. Therefore, 37 distribution functions are utilized to evaluate the wind speed characteristics at the selected locations. The expression of the probability density function (PDF) and cumulative distribution function (CDF) for the chosen distribution models are shown in Table S2 as supplementary material. Generally, the distribution parameters can be estimated using various methods such as moments, maximum likelihood estimation, and least-squares methods (Allouhi et al. 2017). In the present study, the parameters of the selected distribution models were calculated using the maximum-likelihood approach. For this purpose, Easyfit software was utilized.

Goodness-of-fit test

The expression of the Kolmogorov–Smirnov (K-S) test, the Anderson–Darling (A-D) test, and the Chi-squared (C-s) test are given in Eqs. (1)–(3). These tests were used to find a suitable model for analyzing the wind speed distribution.

$$D = \max_{1\le i\le n}\left(F\left(x_i\right) - \frac{i-1}{n}, \frac{i}{n} - F\left(x_i\right)\right) \quad (1)$$

Kolmogorov–Smirnov test
where:

$$F_n(x) = \frac{1}{n} \times (Numberofobservation \le x) \quad (2)$$

$$A^2 = -n - \frac{1}{n}\sum_{i=1}^{n}(2i-1)\times\left[lnF_X\left(x_i\right) + ln\left(1 - F_X\left(x_{n-i+1}\right)\right)\right] \quad (3)$$

Anderson–Darling test

where:$F_X\left(x_i\right)$ is the cumulative distribution function of the proposed distribution atx_i, for $i = 1, 2, \ldots, n$.

$$\mathcal{X}^2 = \sum_{i=1}^{k}\frac{\left(O_i - E_i\right)^2}{E_i} \quad (4)$$

Chi-squared test
where O_i is the observed frequency for bin i, and E_i is the expected frequency for bin i calculated by:

$$E_i = F\left(x_2\right) - F\left(x_1\right) \quad (5)$$

where F is the cumulative distribution function of the probability distribution being tested, and x_1 and x_2 are the limits for bin i.

Estimation of the wind turbine output

The wind power density (WPD) is calculated to assess wind power potential for designing the wind farm. Equations (6) and (7) are utilized to determine the WPD (Ayodele et al. 2014; Mohammadi et al. 2017).

$$\frac{P}{A} = \frac{1}{2}\rho v^3 \quad (6)$$

$$\frac{P}{A} = \frac{1}{2}\rho v^3 f(v) \quad (7)$$

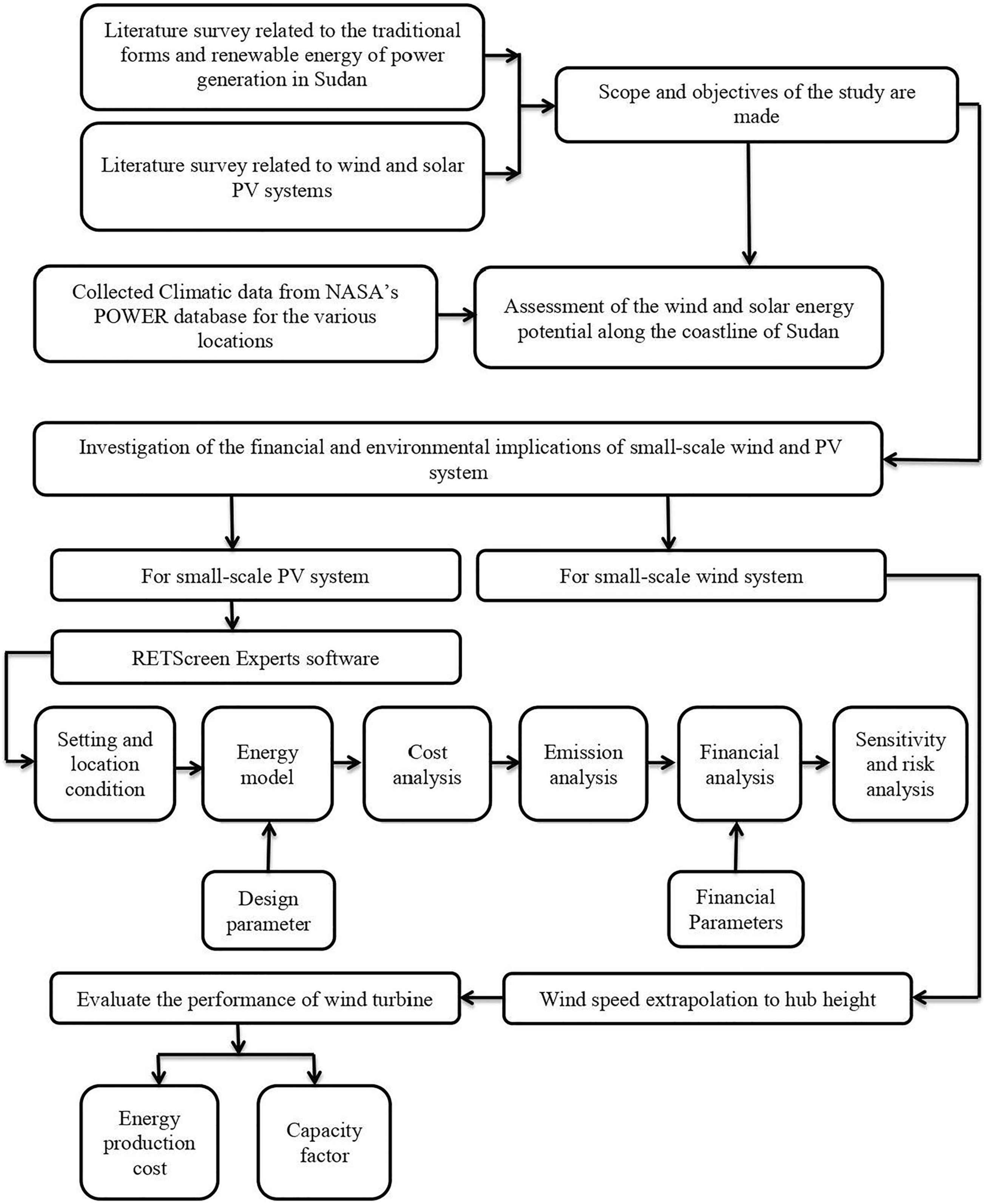

Fig. 1 Schematic description for the proposed methodology

Fig. 2 Map of Sudan (selected region▭)

Table 1 Geographical coordinates and altitude of 17 locations

Location number	Name of location	Latitude (° N)	Longitude (° E)	Elevation (m)
Location 1	Bi'r Shalatayn	23.040	35.508	42
Location 2	Aalaba Nature Reserve	22.998	35.565	28
Location 3	Bi'r al Hasa	22.960	35.660	2
Location 4	Sudanese Halayeb petrol	22.614	35.952	22
Location 5	Halaib Sudan	22.430	36.341	28
Location 6	Abdo Coffee	22.401	36.420	1
Location 7	Halayeb	22.220	36.641	11
Location 8	Fudukwan	21.750	36.738	216
Location 9	Marsa Oseif	21.768	36.862	15
Location 10	Dungunab	21.1058	37.1192	9
Location 11	Muhammad Qol	20.9052	37.1587	4
Location 12	Arb'at	19.8667	37.1833	21
Location 13	Portsudan	19.5903	37.1902	13
Location 14	Suakin	19.1040	37.3293	2
Location 15	Trinkitat	18.6833	37.7167	5
Location 16	Tokar	18.4260	37.7275	24
Location 17	Afflanda	18.0500	38.4167	10

Furthermore, the average WPD can be determined using Eq. (8) (Irwanto et al. 2014).

$$\frac{\overline{P}}{A} = \frac{1}{2}\rho\overline{v}^3 \tag{8}$$

where P is wind power density in W,$\overline{P}$ is mean wind power density in W, A is swept area in m^2, ρ is the air density in kg/m^3, $f(v)$ is the probability density function (PDF), and $\overline{v}$ is the mean wind speed in m/s.

Moreover, the power-law model is used to estimate the wind speed at various hub heights, and it can be expressed as Eq. (9) (Irwanto et al. 2014):

$$\frac{v}{v_{10}} = \left(\frac{z}{z_{10}}\right)^{\alpha} \tag{9}$$

where v is the wind speed at the wind turbine hub height z, v_{10} is the wind speed at the original height z_{10}, and α is the surface roughness coefficient (Eq. (10)) (Irwanto et al. 2014; Masseran 2015).

$$\alpha = \frac{0.37 - 0.088ln\left(v_{10}\right)}{1 - 0.088ln\left(z_{10}/10\right)} \tag{10}$$

Besides, the ratio of average power output (E_{out}) to the rated power of wind turbine (E_r) is called the capacity factor (C_f), and it can be calculated as given below (Ayodele et al. 2014; Pallabazzer 2003; Nouni et al. 2007).

$$C_f = \frac{E_{out}}{E_r} = \frac{Meanoutputpoweroveraspecificperiod}{ratepowerofwindturbine} \tag{11}$$

$$E_{out} = \sum_{i=1}^{n} P_{wt(i)}t \tag{12}$$

$$P_{wt(i)} = \begin{cases} P_r\frac{v_i^2 - v_{ci}^2}{v_r^2 - v_{ci}^2}\left(v_{ci} \le v_i \le v_r\right) \\ \frac{1}{2}\rho A C_p v_r^2\left(v_r \le v_i \le v_{co}\right) \\ 0\left(v_i \le v_{ci} and v_i \ge v_{co}\right) \end{cases} \tag{13}$$

$$E_r = P_r.t \tag{14}$$

$$C_p = 2\frac{P_r}{\rho A v_r^3} \tag{15}$$

where t is the number of hours in the period under consideration, where v_i is the vector of possible wind speed at a given site, $P_{wt(i)}$ is the vector of the corresponding wind turbine output power (W), P_r is the rated power of the turbine (W), v_{ci} is the cut-in wind speed (m/s), v_r is the rated wind speed (m/s), and v_{co} is the cut-out wind speed (m/s) of the wind turbine.

Moreover, the energy production cost (EPC) from the wind turbine (Eq. (16)) can be calculated based on the present value cost (PVC) and capacity factor as shown in Eq. (17) (Ohunakin et al. 2013; Belabes et al. 2015).

$$UCE = \frac{PVC}{E_r \times CF} \tag{16}$$

$$PVC = \left[I + C_{omr}\left(\frac{1+i}{r-i}\right) \times \left[1 - \left(\frac{1+i}{1+r}\right)^n\right] - S\left(\frac{1+i}{1+r}\right)^n\right] \tag{17}$$

Table 2 Descriptive statistics of wind speed data for the investigation period

Location number	Mean	SD	CV	Minimum	Maximum	Skewness	Kurtosis
1	4.6049	0.1805	3.92	4.2854	5.1048	0.58	0.33
2	3.964	0.1151	2.9	3.7759	4.2943	0.96	0.9
3	3.9641	0.1147	2.89	3.7709	4.2957	0.93	0.88
4	4.7299	0.171	3.61	4.4624	5.1653	0.81	0.3
5	2.8818	0.2073	7.19	2.6045	3.3957	0.66	−0.35
6	4.1134	0.1303	3.17	3.9122	4.444	0.91	0.13
7	5.1663	0.2269	4.39	4.768	5.7092	0.86	0.52
8	4.6142	0.2109	4.57	4.2785	5.1831	0.97	0.69
9	4.8441	0.2247	4.64	4.5117	5.4086	0.92	0.31
10	4.4569	0.1929	4.33	4.1887	4.9288	0.84	0.04
11	4.1524	0.1505	3.62	3.9144	4.4964	0.66	−0.17
12	3.9672	0.1035	2.61	3.7262	4.1601	0.14	−0.16
13	3.9964	0.0889	2.23	3.7498	4.1212	−0.68	−0.16
14	4.2882	0.1305	3.04	4.0387	4.5212	0.02	−1.02
15	4.4097	0.1088	2.47	4.1452	4.6134	−0.32	−0.14
16	4.6621	0.171	3.67	4.2737	4.9758	−0.06	−0.61
17	4.9694	0.1616	3.25	4.6413	5.2729	−0.15	−0.78

where r is the discount rate; i is the inflation rate; n is the machine life as designed by the manufacturer; C_{omr} is the cost of operation and maintenance; I is the investment summation of turbine price and other initial costs, including provisions for civil work, land, infrastructure, installation, and grid integration; and S is the scrap value of the turbine price and civil work.

Solar energy analysis procedure

This work develops grid-connected PV systems for electricity generation for buildings in the selected location. In the literature, power generating factors, energy demand, solar PV energy required, PV module sizing, and inverter sizing are essential factors for designing a PV system (Kassem et al. 2021a; Kassem et al. 2020b; Owolabi et al. 2019).

$$Powergeneratingfactor = \frac{solarirradiance \times sunshinehour}{standardtestconditionirradiance} \tag{18}$$

$$Energydemand = Energyconsumptionofallloads \tag{19}$$

$$SolarPVenergyrequired = peakenergyrequirement \times energylostinthesystem \tag{20}$$

$$totalwattpeakratting = \frac{SolarPVenergy}{generationofpanelneededfactor} \tag{21}$$

$$PVmodulesize = \frac{totalwattpeakratting}{PVoutputpowerrating} \tag{22}$$

$$Invertersizing = peakenergyrequirement \times 1.3 \tag{23}$$

Simulation tool for investigation of the PV system's feasibility

In general, several software (RETScreen, HOMER, PVsyst, TRANSYS, and so on) are used to estimate the performance of the renewable energy systems (Markovic et al. 2011). In the last few years, RETScreen has been widely utilized to estimate the performance of various renewable energy systems as well as the feasibility of these systems (Sinha and Chandel 2014). Natural Resources Canada (NRC) developed RETScreen software. It helps estimate energy production, capacity factor, emissions reduction, and financial viability. The software uses the long-term NASA monthly average meteorological data. Numerous scientific researches have investigated the feasibility of renewable energy systems such as wind farms and PV plants (Kebede 2015; Rafique et al. 2018; Datta et al. 2020, Kassem et al. 2021c). For instance, Kebede (2015) explored the potential of a 5-MW grid-connected PV system using RETScreen software in Ethiopia. Rafique et al. (2018) used RETScreen software to evaluate

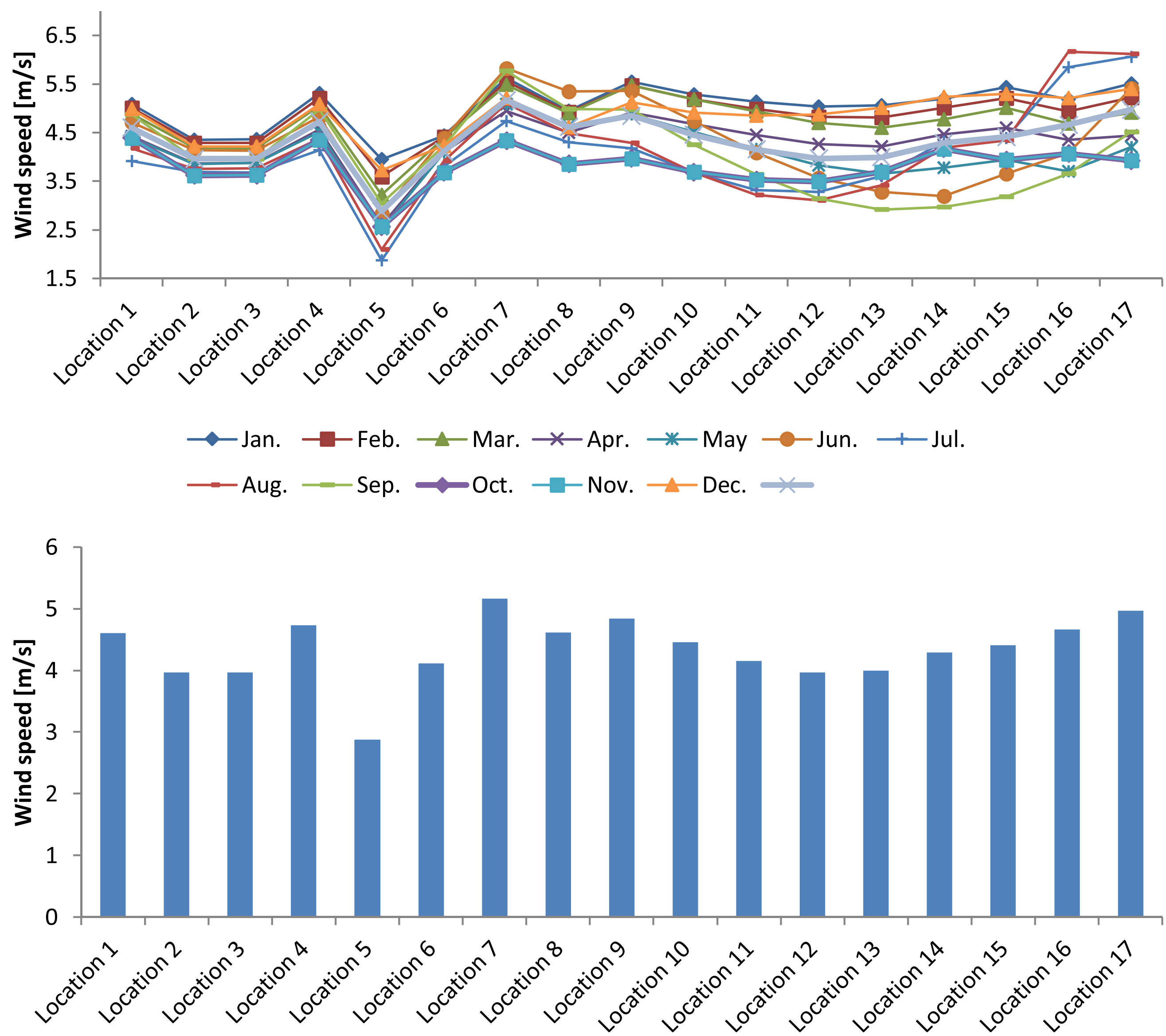

Fig. 3 Average wind speed: (**a**) annual wind speed and (**b**) monthly wind speed data

the feasibility of a 100-MW grid-connected wind farm in different locations in Saudi Arabia. Datta et al. (2020) utilized RETScreen software to investigate the techno-economic feasibility of solar rooftop PV system Bangladesh. Kassem et al. (2021c) investigated the effect of orientation angles on the techno-economic feasibility of 1789 kW grid-connected PV systems in Near East University Hospital, Northern Cyprus, using RETScreen software. Consequently, RETScreen software was utilized to determine economic feasibility indicators, which are used to assess the performance of the proposed systems. The mathematical expressions of these indicators are given below.

Capacity factor (CF)

$$CF = \frac{P_{out}}{P \times 8760} \tag{24}$$

Annual GHG emission reduction (A-GHG)

$$A - GHG = \left[(BasecaseGHGemissionfactor) - (ProposedcaseGHGemissionfactor) \right] \times Enduseenergydelivered \tag{25}$$

GHG emission reduction cost (GHG-E-RC)

$$\mathrm{GHG - E - RC} = \frac{ALCS}{\Delta_{GHG}} \tag{26}$$

Net present value (NPV)

$$NPV = \sum_{n=0}^{N} \frac{C_n}{(1+r)^n} \tag{27}$$

Levelized cost of energy (LCOE)

$$LCOE = \frac{sumofcostoverlifetime}{sofelectricitygeneratedoverthelifetime} \tag{28}$$

Simple payback (SP)

$$SP = \frac{C - IG}{\left(C_{ener} + C_{capa} + C_{RE} + C_{GHG}\right) - \left(C_{o\&M} + C_{fuel}\right)} \tag{29}$$

Equity payback (EP)

$$EP = \sum_{n=0}^{N} C_n \tag{30}$$

Annual life cycle savings (ALCS)

$$ALCS = \frac{NPV}{\frac{1}{r}\left(1 - \frac{1}{(1+r)^N}\right)} \tag{31}$$

Benefit–Cost ratio (B-C)

$$B - C = \frac{NPV + \left(1 - f_d\right)C}{\left(1 - f_d\right)C} \tag{32}$$

where P_{out} is energy generated per year,P is installed capacity, N is the project life in years, C_n is the after-tax cash flow in year n, and r is the discount rate. C is the total initial cost of the project, f_d is the debt ratio, B is the total benefit of the project, IG is the incentives and grants, and C_{ener} is the annual energy savings or income. C_{capa} is the annual capacity savings or income, C_{RE} is the annual renewable energy (RE) production credit income, C_{GHG} is the GHG reduction income, and $C_{o\&M}$ is the yearly operation and maintenance costs incurred by the clean energy project. C_{fuel} is the annual cost of fuel, which is zero for renewable projects, and Δ_{GHG} is the annual GHG emission reduction.

Results and discussion

Evaluation of solar and wind potential energy resources in Sudan

Wind energy characteristics

The descriptive statistics of wind speed data for each location are presented in Table 2. It is observed that the mean wind speed is within the range of 2.88–5.17 m/s. It is found that locations 7 and 5 have the highest and lowest wind speed value, respectively. Additionally, it is noticed that the values of coefficient of variation are within the range of 2.23–7.19, as shown in Table 2.

Moreover, it is observed that the lowest and highest wind speed values are recorded in location 11 and location 5, with a value of 1.87 m/s and 6.16 m/s, respectively, as shown in Fig. 3. In addition, it is observed that location 5 has the highest annual value of wind speed (i.e., 5.17 m/s) compared to other areas, as shown in Fig. 3.

As mentioned previously, the parameters of the selected distribution models were calculated using the maximum-likelihood approach. In addition, K-S, A-D, and C-s tests were used to find the best distribution function. Generally, the minimum value of K-S, A-D, and C-s tests will define the best distribution that fits the wind speed data. The used models' parameters and K-S, A-D, and C-s tests are summarized in Table S3–S19 as supplementary material. Besides, Table 3 summarizes the best distribution functions that gave the best fit to the actual data. Based on the findings, it is observed that Wakeby distribution has the lowest value of K-S, A-D, and C-s, which is considered as the best distribution function to analyze the characteristics of wind speed for all selected locations. The PDF and CDF plots for all distribution models are illustrated in Figure S1–S17 as supplementary material.

According to the previous studies, the air density value is assumed to be 1.23 kg/m^3. Hence, the average value of WPD is determined by using Eq. (8). Table 3 summarizes

Table 3 The best model (Rank #1) for studying the characteristics of wind speed

Location	Distribution function	Goodness of fit			Location	Distribution function	Goodness of fit		
		K-S	**A-D**	**C-S**			**K-S**	**A-D**	**C-S**
1	Gen. Pareto	-	0.3256	-	9	Wakeby	0.13104	-	-
	Wakeby	0.10717	-	-		Wakeby	-	0.34034	-
	Weibull	-	-	2.85E-06		Weibull	-	-	4.14E-06
2	Wakeby	0.12866	-	-	10	Wakeby	0.17057	-	-
	Wakeby	-	0.26023	-		Wakeby	-	0.46282	-
	Pearson 5	-	-	9.77E-06		Burr	-	-	0.01582
3	Wakeby	-	0.2426	-	11	Wakeby	0.11742	-	-
	Wakeby	0.2426	-	-		Wakeby	-	0.31179	-
	Longnormal	-	-	4.6E-06		Beta	-	-	1.40E-04
4	Beta	-	-	9.39E-06	12	Wakeby	0.13715	-	-
	Wakeby	-	0.2567	-		Wakeby	-	0.36497	-
	Dagum (4P)	0.13964	-	-		Pareto	-	-	0.02431
5	Gen.Extreme	-	0.23376	-	13	Log-Logistic	0.16815	-	-
	Weibull (3P)	0.14415	-	-		Wakeby	-	0.35794	-
	Log-Logistic	-	-	2.20E-04		Exponential	-	-	0.06183
6	Wakeby	0.13024	-	-	14	Cauchy	0.13568	-	-
	Wakeby	-	0.31623	-		Gen. Extreme	-	0.24203	-
	Gumbel Min	-	-	4.84E-04		Weibull	-	-	0.00249
7	Gen.Extreme	0.09977	-	-	15	Log-Gamma	0.13893	-	-
	Gen. Extreme	-	0.2033	-		Gen. Extreme	-	0.25506	-
	Rayleigh	-	-	5.96E-04		Weibull	-	-	0.00847
8	Gumbel Min	0.13513	-	-	16	Log-Logistic	0.13467	-	-
	Wakeby	-	0.24862	-		Wakeby	-	0.21704	-
	Beta	-	-	2.79E-08		Beta	-	-	7.52E-08
9	Wakeby	0.13104	-	-	17	Gen.Extreme	0.12468	-	-
	Wakeby	-	0.34034	-		Wakeby	-	0.2546	-
	Weibull	-	-	4.14E-06		Beta	-	-	1.71E-05
10	Wakeby	0.17057	-	-					
	Wakeby	-	0.46282	-					
	Burr	-	-	0.01582					

the value of the mean, SD, CV, and WPD for the best distribution models (Table 4). Based on the results of WPD, ranging from 13.950 (location 5) to 84.459 W/m^2 (location 7), it is concluded that the potential of wind energy in the selected location is categorized as poor (class 1) (Khan et al. 2019). Therefore, a small-scale wind turbine can produce electricity from the wind energy for the buildings.

Solar energy characteristics

Generally, solar radiation, including global horizontal irradiation (GHI) and direct normal irradiation (DNI), and air temperature (AT), are essential factors for evaluating the performance of PV systems. Table 5 shows the categorization of the solar resource at a specific location based on the value of GHI and DNI.

Tables S20 and S21 as supplementary material summarize the average monthly value of GHI and DNI for all selected locations. Moreover, the annual values of GHI are varied from 2189 to 2421 kWh/m^2, as shown in Fig. 4. It is observed that locations 2 and 16 have the maximum and minimum value of GHI and DNI with 2412 kWh/m^2 and 2594 kWh/m^2 and 2189 and 1943 kWh/m^2, respectively (see Fig. 4). Moreover, it is found that the solar resources in the selected locations are classified as excellent, outstanding, or superb potential classes, as shown in Tables S20 and S21 as supplementary material. The solar resource in location 2 and location 7 was classified as superb (class 7).

Consequently, it is concluded that the selected locations are suitable for installing large-scale or small-scale photovoltaic systems due to the high value of GHI and DNI. Accordingly, all chosen regions are appropriate for installing

Table 4 Parameter values of the best distribution functions over the investigated period at 10 m height

Locations	Distribution functions	Parameters					
		Mean (m/s)	SD (m/s)	CV	Skewness	Kurtosis	WPD (W/m^2)
Location1	Actual	4.605	0.181	3.920	0.580	0.330	59.809
	Wakeby	-	-	-	-	-	-
	Gen. Pareto	4.605	0.379	0.082	− 0.287	− 1.173	59.809
	Weibull	4.544	0.445	0.098	− 0.723	0.802	57.448
Location 2	Actual	3.964	0.115	2.900	0.960	0.900	38.151
	Wakeby	3.964	0.275	0.069	0.093	− 1.163	38.151
	Wakeby	3.964	0.275	0.069	0.093	− 1.163	38.151
	Pearson 5	3.964	0.257	0.065	0.260	0.127	38.151
Location 3	Actual	3.964	0.115	2.890	0.930	0.880	38.154
	Wakeby	3.964	0.272	0.069	0.130	− 1.142	38.154
	Wakeby	3.964	0.272	0.069	0.130	− 1.142	38.154
	Longnormal	3.964	0.252	0.064	0.191	0.065	38.154
Location 4	Actual	4.730	0.171	3.610	0.810	0.300	64.813
	Beta	4.730	0.393	0.083	− 0.009	− 1.429	64.813
	Wakeby	4.730	0.404	0.085	0.090	− 1.164	64.813
	Dagum (4P)	4.758	0.466	0.098	1.673	12.553	65.967
Location 5	Actual	2.882	0.207	7.190	0.660	− 0.350	14.659
	Weibull (3P)	2.882	0.615	0.213	0.534	0.080	14.654
	Gen.Extreme	2.882	0.676	0.235	0.472	0.203	14.659
	Log-Logistic	2.835	0.812	0.286	1.569	8.526	13.950
Location 6	Actual	4.113	0.130	3.170	0.910	0.130	42.629
	Wakeby	4.113	0.293	0.071	− 0.495	− 1.023	42.629
	Wakeby	4.113	0.293	0.071	− 0.495	− 1.023	42.629
	Gumbel Min	4.113	0.283	0.069	− 1.140	2.400	42.629
Location 7	Actual	5.166	0.227	4.390	0.860	0.520	84.459
	Gen.Extreme	5.166	0.540	0.104	− 0.667	0.315	84.459
	Gen. Extreme	5.166	0.540	0.104	− 0.667	0.315	84.459
	Rayleigh	5.166	2.701	0.523	0.631	0.245	84.459
Location 8	Actual	4.614	0.211	4.570	0.970	0.690	60.172
	Gumbel Min	4.614	0.451	0.098	− 1.140	2.400	60.172
	Wakeby	-	-	-	-	-	-
	Beta	4.614	0.451	0.098	− 0.019	− 1.266	60.172
Location 9	Actual	4.844	0.225	4.640	0.920	0.310	69.622
	Wakeby	4.844	0.614	0.127	− 0.428	− 1.083	69.622
	Wakeby	4.844	0.614	0.127	− 0.428	− 1.083	69.622
	Weibull	4.760	0.720	0.151	− 0.523	0.306	66.050
Location 10	Actual	4.457	0.193	4.330	0.840	0.040	54.226
	Wakeby	4.457	0.650	0.146	− 0.072	− 1.213	54.226
	Wakeby	4.457	0.650	0.146	− 0.072	− 1.213	54.226
	Burr	4.466	0.617	0.138	− 0.569	0.408	54.558
Location 11	Actual	4.152	0.151	3.620	0.660	− 0.170	43.853
	Wakeby	4.152	0.722	0.174	0.126	− 1.144	43.853
	Wakeby	4.152	0.722	0.174	0.126	− 1.144	43.853
	Beta	4.152	0.703	0.169	0.050	− 1.553	43.853
Location 12	Actual	3.967	0.104	2.610	0.140	− 0.160	38.244
	Wakeby	3.967	0.747	0.188	0.455	− 0.790	38.244
	Wakeby	3.967	0.747	0.188	0.455	− 0.790	38.244
	Pareto	4.032	1.264	0.313	5.841	217.230	40.148

Table 4 (continued)

Locations	Distribution functions	Parameters					
		Mean (m/s)	SD (m/s)	CV	Skewness	Kurtosis	WPD (W/m^2)
Location 13	Actual	3.996	0.089	2.230	−0.680	−0.160	39.094
	Log-Logistic	3.944	0.901	0.228	1.190	5.089	37.577
	Wakeby	3.996	0.740	0.185	0.391	−0.883	39.094
	Exponential	3.996	3.996	1.000	2.000	6.000	39.094
Location 14	Actual	4.288	0.131	3.040	0.020	−1.020	48.298
	Cauchy	-	-	-	-	-	-
	Gen. Extreme	4.288	0.752	0.175	−0.601	0.190	48.298
	Weibull	4.187	0.849	0.203	−0.343	−0.009	44.962
Location 15	Actual	4.410	0.109	2.470	−0.320	−0.140	52.521
	Log-Gamma	4.416	0.751	0.170	0.763	1.101	52.761
	Gen. Extreme	4.410	0.745	0.169	0.000	−0.283	52.521
	Weibull	4.299	0.796	0.185	−0.403	0.083	48.657
Location 16	Actual	4.662	0.171	3.670	−0.060	−0.610	62.066
	Log-Logistic	4.567	0.930	0.204	1.043	4.111	58.352
	Wakeby	3.496	0.856	0.184	0.681	−0.361	26.162
	Beta	4.662	0.820	0.176	0.367	−1.255	62.066
Location 17	Actual	4.969	0.162	3.250	−0.150	−0.780	75.165
	Gen.Extreme	4.969	0.804	0.162	0.023	−0.277	75.165
	Wakeby	4.969	0.790	0.159	0.026	−1.192	75.165
	Beta	4.969	0.768	0.155	0.089	−1.483	75.165

Table 5 Classification of solar energy (Prăvălie et al. 2019)

Class	Annual GHI (kWh/m^2)	Class	Annual DNI (kWh/m^2)
1 (Poor)	< 1191.8	1 (Poor)	< 936.9
2 (marginal)	1191.8–1419.7	2 (marginal)	936.9–1255.7
3 (fair)	1419.7–1641.8	3 (fair)	1255.7–1546.8
4 (good)	1641.8–1843.8	4 (good)	1546.8–1840.9
5 (excellent)	1843.8–2035.9	5 (execllent)	1840.9–2149.9
6 (outstanding)	2035.9–2221.8	6 (outstanding)	2149.9–2533.7
7 (superb)	> 2221.8	7 (superb)	> 2533.7

PV/flat-plate and concentrated solar power (CSP) systems. Furthermore, it is noticed that the maximum and minimum average value of AT values was recorded in location 17 and location 1 with a value of 32.47 and 28.35 °C, respectively, as shown in Fig. 4 and Table S22 as supplementary material.

Summary

The previous sections give the following findings:

- The value of wind power density (13.950–84.459WW/m^2) indicated that the low cut-in wind turbine such as vertical axis wind turbine could be considered a good solution for generation electricity from wind energy in the selected locations.
- The selected locations are suitable for installing large-scale or small-scale photovoltaic systems due to the high value of GHI and DNI. Accordingly, all chosen regions are appropriate for installing PV/flat-plate and concentrated solar power (CSP) systems.

Economic analysis

This section discusses the economic validity of the wind and solar systems. It should be noted that the cost of both systems has been estimated to be around USD 3300, which

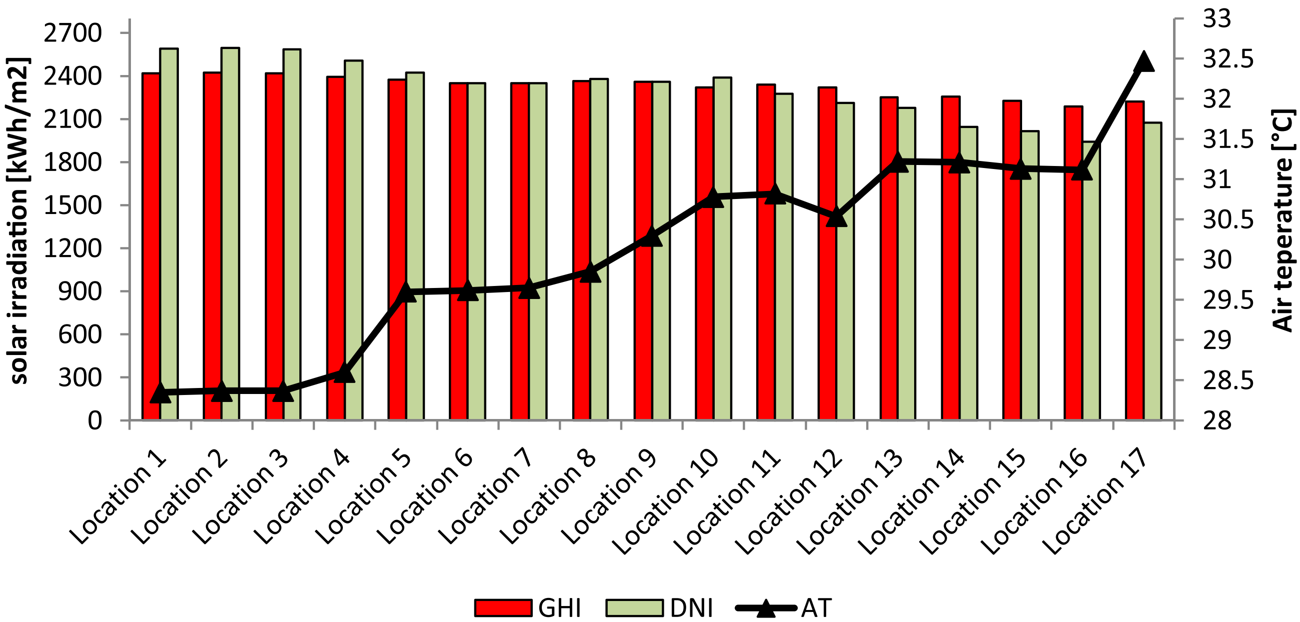

Fig. 4 The annual value of GHI and DNI as well as average AT for all selected locations

Table 6 Specifications of small-scale vertical wind turbines

Model No	Model	Rated power (kW)	Rated wind speed (m/s)	Cut-in wind speed (m/s)
#1	WS-4B & 4C 1–2 kW	1 kW	18 m/s	2 m/s
#2	3000 W Vertical Axis Wind Turbine	3kw	9.0 m/s	1.0 m/s
#3	10kw H-Type Vertical Axis Wind	10kw	12 m/s	2.5 m/s
#4	1kw AC Three-Phase Vertical	1kw	12 m/s	2 m/s
#5	2kw AC Three-Phase Vertical	2kw	12 m/s	2 m/s
#6	3 kW AC Three-Phase Vertical	3kw	12 m/s	2 m/s
#7	5kw AC Three-Phase Vertical	5kw	12 m/s	2 m/s
#8	10 kW AC Three-Phase Vertical	10kw	12 m/s	2.5 m/s
#9	Vertical Axis Wind Generator V	5KW	10 m/s	2.5 m/s
#10	AC-120 V 5kw Vertical Axis Wind	5 kw	12 m/s	2 m/s

is determined based on the recent market data in the country and cost prices available in the literature. Furthermore, the economic and environmental elements are estimated to assess the effectiveness of the aforementioned proposed systems. Financial metrics such as inflation rate (10.13%), discount rate (12.68%), reinvestment rate (16.01%), debt ratio (70%), and debt interest rate (5.88%) are used as input variables in this study.

Economic analysis of wind turbine

As mentioned previously, small-scale wind turbines, mainly vertical axis wind turbines, are suitable to be used for generating power from wind energy in the selected locations. The selected wind turbine has been chosen after an overall comparison between different types of wind turbines in the current paper. Ten small vertical axis wind turbines have a choice in this study. The characteristics of the selected wind turbines are summarized in Table 6. Generally, the household/building with a various number of floors (2–6 floors) is within the range of 400–600m^2. Thus, it was assumed that the height of the building is about 12 m (the approximate number of the floor is three floors).

Figure 5 illustrates the annual output energy, the capacity factor, and energy unit cost per kWh based on the PVC method for all models. It is found that the values of annual energy output are within the range of 158.50–29,063.93kWh; i.e., the wind turbine model of #9 and #1, respectively, achieves the highest and lowest values of annual energy

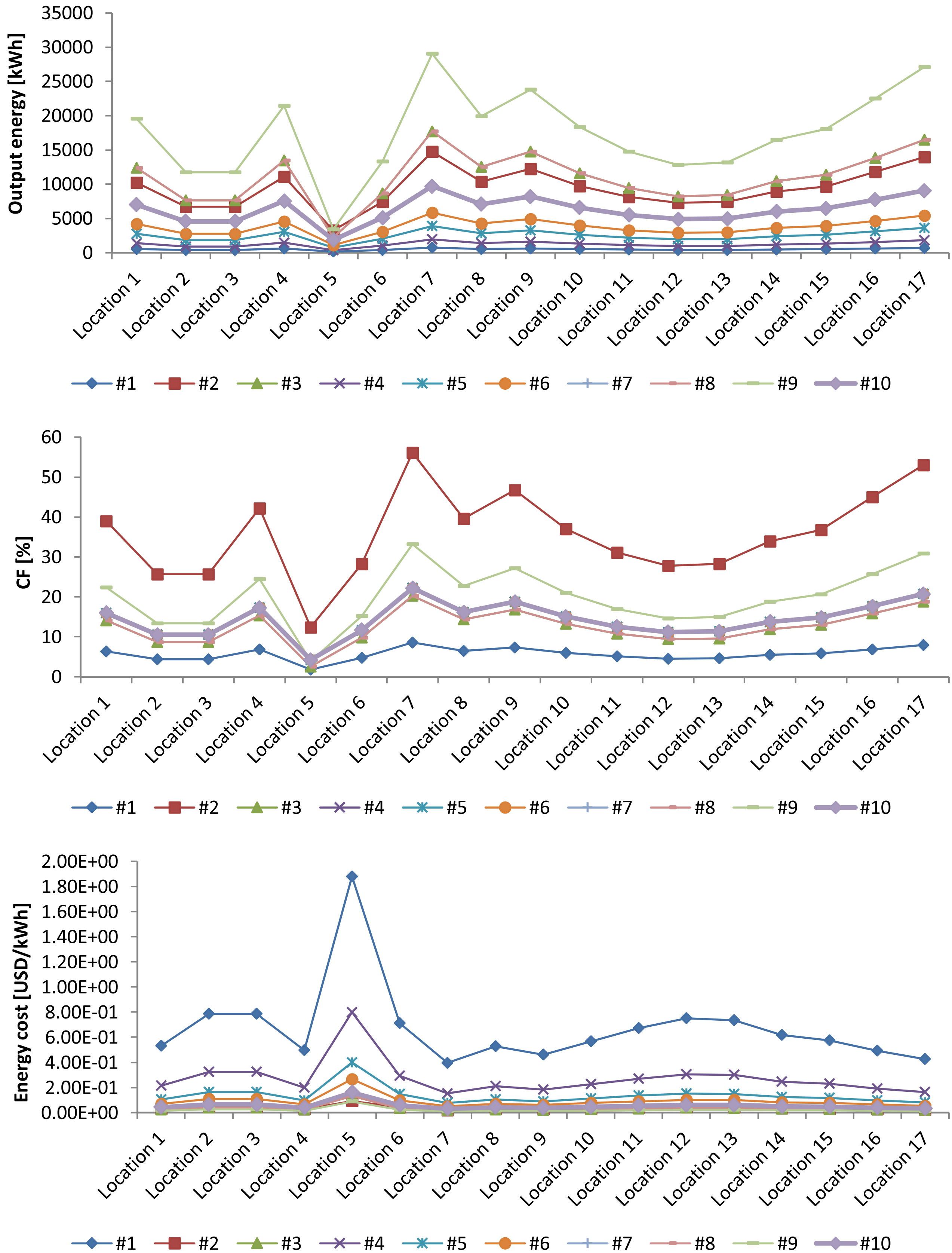

Fig. 5 Annual output energy, the capacity factor, and energy production cost for all locations

Table 7 PV module specification

Item	Specification
Manufacturer	Suntech Power (China)
Model	STPXXXS-B72/Vnh
Nominal power (W)	450 W
Open-circuit voltage (V)	49.2 V
Short-circuit current (A)	11.61 A
The voltage at point of maximum power (V)	1500 V DC (IEC)
Current at point of maximum power (A)	20 A
Module area (m^2)	166
Efficiency (%)	20.7%
Warranty (Year)	25 year
Cost (USD)	210

Table 8 Specification of the selected inverter

PV module technology	Value
Rated power (W)	6000
Min PPT voltage (V)	100
Max PPT voltage (V)	550
DC startup voltage (V)	100
DC shutdown voltage (V)	80
Max input voltage (V)	550
Max DC power (W)	6500
Max AC power (W)	5000
Max DC current (A)	30
Warranty (year)	10
Efficiency (%)	97.9

output. Besides, it is observed that the maximum capacity factor is about 56.10% obtained from the #2 model and recorded in location 7, as shown in Fig. 5. In general, estimating the cost of energy ($/kWh) is essential to select the best wind turbine between the different models. In fact, due to the problem of collecting precise information about wind turbine costs thus, the cost of wind turbine models was obtained from the literature. It found that model#9 (Vertical Axis Wind Generator-V) has the lowest cost value compared to the other selected turbines for the studied locations.

Economic analysis and feasibility study of 5 kW grid-connected PV system

For the proposed small-scale grid-connected PV system, SUNTECH-450 Watt was selected, which is manufactured by Suntech Power. It was chosen in this study as it is one of the market's most efficient PV modules. Table 7 summarizes the specification of the selected module. The total number of modules and the area required for the proposed system is 12 modules and 27m^2. Moreover, one inverter is used to convert the DC into AC and feed it directly to the grid with a capacity of 6 kW and 97.9% efficiency for the proposed PV system (see Table 8).

In the literature, Photovoltaic Geographical Information System (PVGIS) simulation tool (Abdallah et al. 2020; Bailek et al. 2018) can provide the optimum orientation angles (tilt and azimuth angle). Thus, optimum slop and azimuth angles were found to be 35 ° and 0 ° by using PVGIS. Moreover, according to Mehmood et al. (2014) and

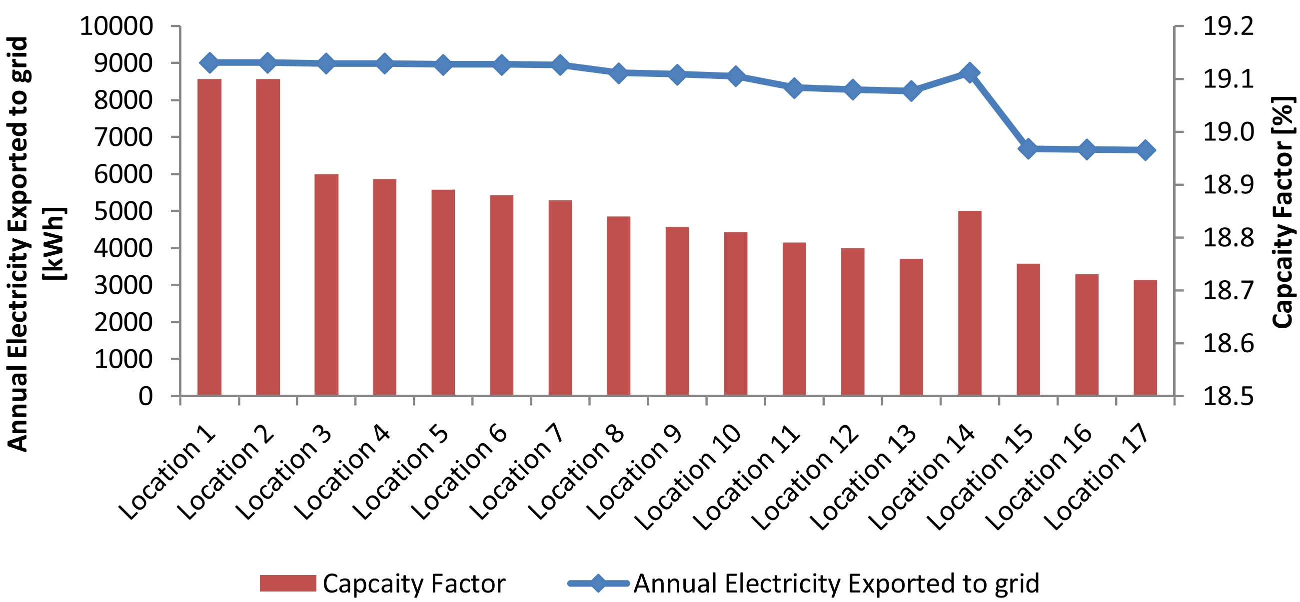

Fig. 6 Annual electricity generation and capacity factor for the developed systems

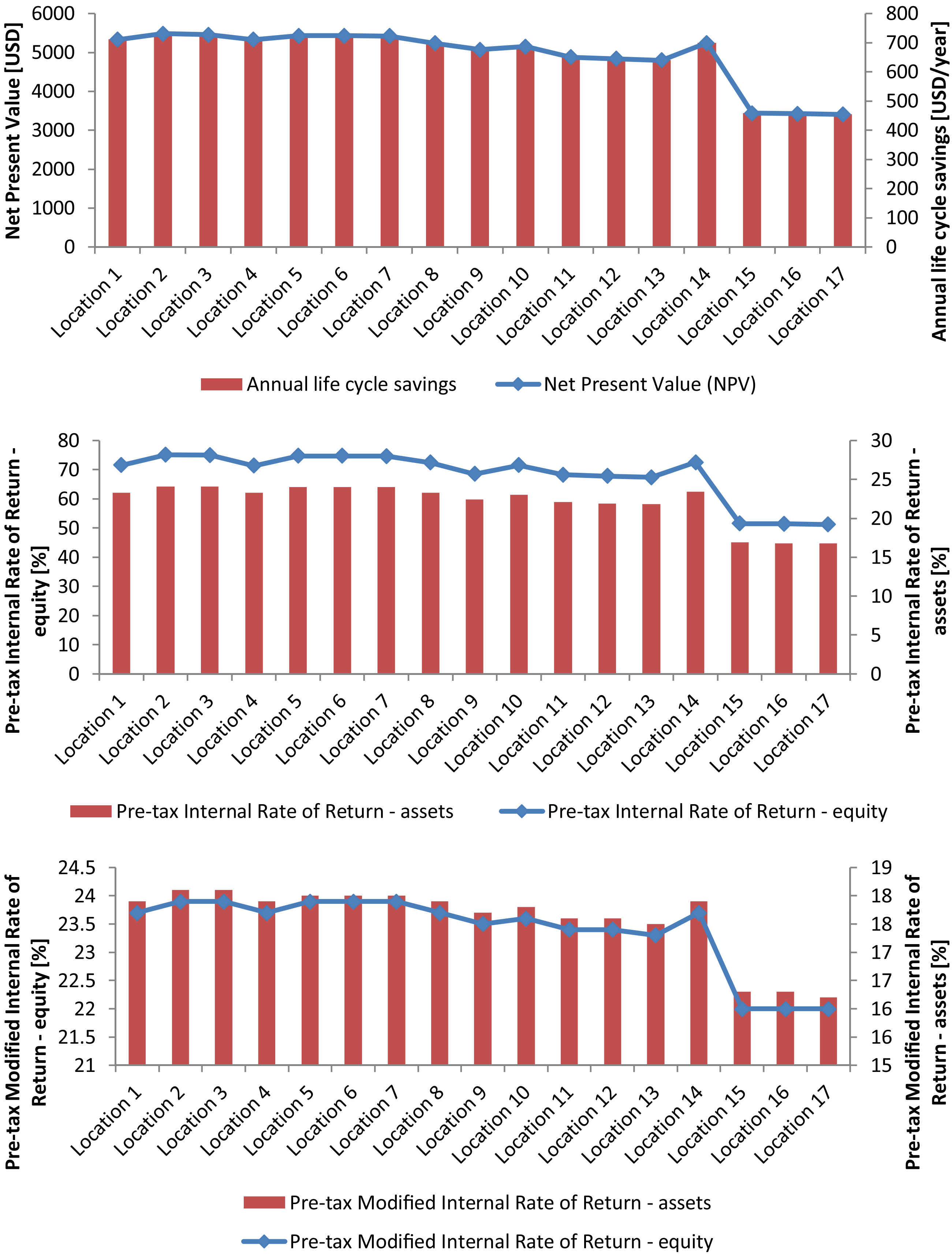

Fig. 7 Performance of 5 kW grid-connected of the solar system

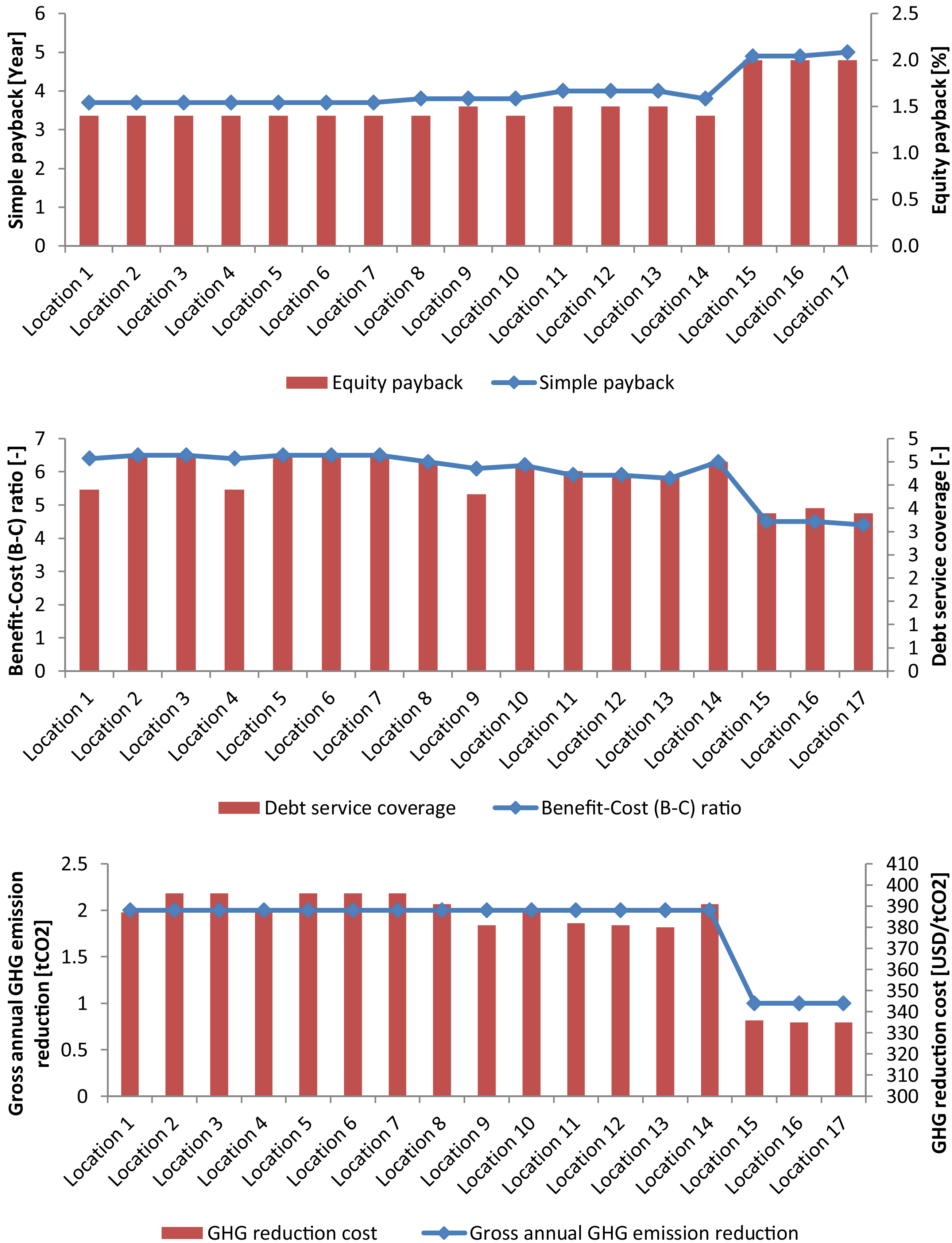

Fig. 7 (continued)

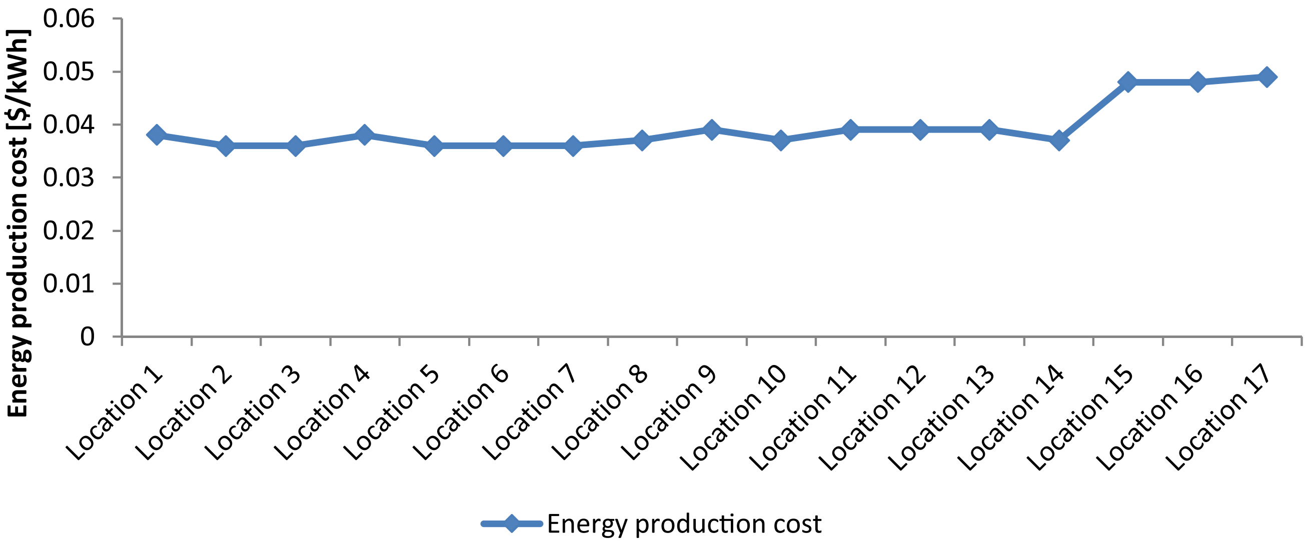

Fig. 7 (continued)

Khandelwal and Shrivastava (2017), the most factors that influence the annual energy and capacity factor of the PV system are solar radiation and the number of clear sunny days. Consequently, the economic feasibility of the developed system was investigated using RETScreen software.

Figure 6 shows the developed systems' annual electricity generation and capacity factor. It is found that the value of EG is within the range of 6648–15,533kWh for the proposed system. The highest value of EG is found in locations 1 and 2 with a value of 9014kWh, while the lowest is found in location 17 with a value of 6648kWh. Furthermore, the CF values are within the range of 18.72–19.1%. Other scientific researchers who investigated the possibility of a grid-connected PV system can back up these findings. For example, Kazem and Khatib (2013) discovered that the projected photovoltaic system in Oman had a CF of 16–23%. Obeng et al. (2020) also found that the CF of grid-connected PV systems using various technologies ranged from 15.37 up to 15.75%. Mohammadi et al. (2018) found that the CF of grid-connected photovoltaic systems with various sun-tracking modes ranged from 17.54 to 27.42%. As a result, the value obtained from the current investigation for each location can be compatible with acceptable values. As a result, installing a grid-connected rooftop PV system throughout the Red Sea state's 17 locations is technically feasible.

Moreover, Fig. 7 illustrates the main results concerning the economic feasibility of the proposed system. The results revealed that the suggested systems had a positive net present value, indicating that the project was viable, as shown in Fig. 7. According to Owolabi et al. (2019) and Mohammadi et al. (2018), it must be fiscally and economically feasible. Furthermore, based on the internal rate of return, which measures a project's profitability, the proposed projects in the selected sites are deemed financially viable (Owolabi et al. 2019; Rehman et al. 2017). As well, the results of the annual life cycle savings (ALCS) are presented in Fig. 7. It was observed that ALCS values are varied from 455 to 729 USD/year. The annual life cycle savings is another term used to get an idea about the benefits of a project (Abd-ur-Rehman and Al-Sulaiman 2016). This term is calculated based on the net present value, project life, and the discount rate.

Furthermore, the equity payback (EP) and simple payback (SP) periods are between 1.4 and 2.0 years and 3.7 and 5.0 years, respectively. These findings suggest that photovoltaic systems in all regions are financially viable (Fig. 7).

Besides, the benefit–cost ratio (B-C) is s a profitability indicator utilized to determine the viability of cash flows generated from a project. It is found that the B-C ratio is within the range of 4.4–6.5, as shown in Fig. 7. Thus, the proposed projects are considered profitable because the values of B-C are greater than one (Abd-ur-Rehman and Al-Sulaiman 2016).

In addition, the emission analysis worksheet is used to measure the reduction of greenhouse gas (GHG) emissions resulting from the solar PV system. In addition, it is utilized to estimate the revenue that may occur from the sales of the GHG reduction emission, as shown in Fig. 7.

Furthermore, the electricity production cost (EPC) from the PV system is varied between 0.036 and 0.049 USD/kWh. Table 9 compares the LCOE value of the planned projects to the typical value of small-scale PV systems in the literature. The EPC values of the suggested systems are determined to be within the range of the literature's highest (3.165$/kWh) and lowest (0.0199$/kWh) LCOE values. It has been observed that the established systems provide a very great

Table 9 Summary of studies on solar tracking systems in different countries based on EPC

Reference	Location	Size/load	System	Tracking mode	EPC ($/kWh)
Adaramola (2015)	Norway	2.07 kW	Grid-connected PV	Fixed Tilt	0.149–11.632
Sinha and Chandel (2016)	India	6 kW	Stand-alone PV with micro WT hybrid power system	Fixed Tilt	2.944
				Two-axis adjustment tracking	3.165
Bahrami et al. (2017)	Nigeria	1 kW	Stand-alone PV	Two-axis continuous tracking mode	0.299
Hammad et al. (2017)	Jordan	7.98 kW	Grid-connected PV	Fixed Tilt	0.097
				Two-axis continuous tracking mode	0.122
Bhakta and Mukherjee (2017)	India	5.19 kWh/d	Stand-alone PV	Fixed Tilt	0.226–0.287
				Two-axis continuous tracking mode	0.523–0.732
Talavera et al. (2019)	Spain	1 kW	Grid-connected PV	Fixed Tilt	0.075
				Two-axis continuous tracking mode	0.109
Fakher Alfahed et al. (2019)	Iraq	4 kW	Stand-alone PV	Fixed Tilt	0.09
Martinopoulos (2020)	Europe	4 kW	Grid-connected PV	Fixed Tilt	0.120–0.390
Tarigan (2020)	Indonesia	3 kWp	Grid-connected PV	Fixed Tilt	0.09
Kassem et al. (2019a, b)	Northern Cyprus	6.4 kW	Grid-connected PV	Fixed Tilt	0.0194–0.0199
Kassem et al. (2020a)	Libya	5 kW	Grid-connected PV	Fixed Tilt	0.0307–0.0270
Çamur et al. (2021)	Lebanon	5 kW	Grid-connected PV	Fixed Tilt	0.0239–0.0246
				Two-axis continuous tracking	0.0243–0.0246
Current study	Sudan	5 kW	Grid-connected PV	Fixed Tilt	0.036–0.049

insight into the project's financial viability for all regions. Moreover, the energy production cost of both wind and solar systems is compared with the current electricity tariffs for Sudan and are tabulated in Table 10.

Conclusions and future work

The potential and validity of grid-connected wind and PV systems in Sudan's Red Sea state were investigated. To calculate the solar and wind energy potential at the selected state, solar radiation, wind speed, and air temperature data were provided by NASA POWER datasets during the period of 1982–2019. For wind energy potential, 37 distribution functions were used to evaluate the wind speed characteristics at the selected locations. The best distribution was selected based on the lowest value of the Kolmogorov–Smirnov (K-S) test, the Anderson–Darling (A-D) test, and the Chi-squared (C-s). It is found that Wakeby is considered the best distribution function to investigate the wind speed characteristics for all selected locations. After that, the value of wind power density was determined. It was observed that the values of wind power density are within the range of 13.950–84.459WW/m^2. These results indicated that the low cut-in wind turbine is considered an excellent option to be utilized to produce electricity in the selected

Table 10 Electricity tariffs in the Sudan and electricity production cost wind and solar systems

Power generation	User type	Tariff (US$c/kWh)	Comments
Grid	Residential and agricultural	1.8	For 0–100kWh
		2.3	For 100–200 kWh
		2.7	Above 200 kWh, for each 3.80kWh
	Industrial	5.4	Flat rate
	Commercial	3.6	Flat rate
Wind system	Residential, agricultural, industrial, and commercial	87.03–10.25	Flat rate and depends on the location
Solar system	Residential, agricultural, industrial, and commercial	36.00–49.00	Flat rate and depends on the location

regions, which can be installed on the rooftop of a building. For solar potential, the annual value of global solar radiation and direct normal radiation was calculated to classify the solar resource in the selected locations. The results demonstrated that the selected locations are suitable for installing large-scale or small-scale photovoltaic systems. Moreover, all chosen regions are appropriate for installing PV/flat-plate and CSP systems.

Moreover, technical electricity generation assessment and economic analysis of wind energy turbines were examined based on the PVC method. Besides, RETScreen software was used to analyze the technical, financial, economic, and environmental analysis of the PV system. The annual electrical energy from wind turbines and solar power is within the range of 158.50–29,063.93kWh and 6648–15,533 kWh, respectively. This amount of energy output would contribute significantly to reducing the effect of global warming and enhancing the sustainable technological development of the country. Additionally, the average energy production cost was within the range of 0.036–0.049 $/kWh for PV systems and 0.08703 to 0.01025 $/kWh for model #9 (Vertical Axis Wind Generator-V). In general, the energy production cost of the proposed systems is higher than the electricity company tariff. However, the utilization of small-scale renewable energy systems will help reduce the dependency on fossil fuel, the effect of global warming, and enhance the sustainable technological development of the country.

Finally, the results of this paper demonstrate that a small-scale grid-connected rooftop wind and PV system in the Red Sea state can solve the electricity crisis and reduce the consumption of fossil fuels and environmental pollution by minimizing the emission of CO_2. The results of this research can help investors in the energy and building sectors and accelerate an informed transition towards a more sustainable future. In this study, it is essential to acknowledge the limitations of this work. First, the financial parameters were assumed based on historical values in the literature. Second, the influence of various parameters such as dust, irradiation intensity, air temperature, and relative humidity was neglected due to the limitation of RETScreen software. There was no consideration of land availability in the present study. Future studies will determine whether the land is available for massive grid-connected PV systems. Future studies should also examine the impact of financial parameters, such as discount rates and inflation rates, on investment. Furthermore, the interaction between the distribution grid and the PV system should be examined to understand the impact of grid-connected PV systems on the distribution grid.

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Author contribution All the authors contributed to the work. MA. H. A. A. collected the data and analyzed them. YK wrote the paper. All authors have read and agreed to the published version of the manuscript.

Data availability The manuscript has no associated data to provide. However, additional information has been provided as supplementary material.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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